

Jannatul Ferdous Srabonee

(628) 268-9404 • jfsrabonee@hotmail.com
LinkedIn • Google Scholar • ResearchGate • Website

As a PhD candidate in Electrical Engineering and Computer Science at UC Merced, my research focuses on designing and developing inclusive input and interaction techniques for digital systems. My current work spans brain-computer interfaces, applied artificial intelligence, gesture mapping, and eye tracking. I am especially interested in extending this work toward adaptive AR/VR interfaces that adjust to a user's perceptual and physical needs.

Education

Ph.D. | Electrical Engineering and Computer Science

2023 – Present

University of California, Merced

Advisor: Dr. Ahmed Arif

Dissertation: Novel Input and Interaction Techniques for Motor and Vision Impaired Individuals

B.Sc. | Computer Science and Engineering

2016 – 2020

Khulna University of Engineering and Technology, Khulna, Bangladesh

Advisor: Dr. MAH Akhand

Thesis: EEG based Alcoholism Detection Using CNN through Correlation- based transformation of Channel's Data

Research Experience

Graduate Student Researcher

2023 – Present

Inclusive Interaction Lab, UC Merced

Research Engineer

2021 – 2023

Advanced Intelligent Multidisciplinary Systems Laboratory, Dhaka, Bangladesh

Undergraduate Researcher

2019 – 2020

Intelligence Systems Lab KUET, Khulna, Bangladesh

Publications & Manuscripts

Conference Papers

- [C.3] BezelBraille: A Braille-Inspired Bezel-Guided Virtual Keyboard for Low Vision and Blind Users. Isna Alfi Bustoni, **Jannatul Ferdous Srabonee**, Gulnar Rakhmetulla, Ahmed Arif. MobileHCI 2026 [Accepted, not published yet]
- [C.2] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription. **Jannatul Ferdous Srabonee**, Ohoud Mosa Alharbi, Wolfgang Stuerzlinger, and Ahmed Arif. CHI 2025 [[Paper](#)][[Presentation](#)]
- [C.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data. Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique. IEEE TENSYP 2020 [[Paper](#)]

Journal Articles

- [J.2] Computer Vision-Based IoT Architecture for Post COVID-19 Preventive Measures. Ahsanul Akib, Kamruddin Nur, Suman Saha, **Jannatul Ferdous Srabonee**, M. F. Mridha.

Journal of Advances in Information Technology 2023 [\[Article\]](#)

- [J.1] Autism Detection from 2D Transformed EEG Signal using Convolutional Neural Network. Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique.
Journal of Computer Science 2022 [\[Article\]](#)

Book Chapter

- [B.1] Alcoholism Detection from 2D Transformed EEG Signal. **Jannatul Ferdous Srabonee**, Zahrul Jannat Peya, MAH Akhand, Nazmul Siddique
Proceedings of International Joint Conference on Advances in Computational Intelligence: IJCACI 2020. Springer. [\[Chapter\]](#)

Under Review

- [2] Gaze & Gape: Hands-Free Text Entry Using Gaze and Mouth-Open Gestures
[1] GazeDriver: Semi and Fully Autonomous Telepresence Robot Control Using Eye Gaze in Virtual Reality

Posters & Presentations

- [Po.2] "AI in Accessibility" | Research Week | 2026 | University of California, Merced
[Po.1] "EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis" | EBRAINS Workshop: Brain Activity across Scales and Species: analysis of experiments and Simulations, | 2022 | Rome, Italy [\[Virtual\]](#)
[Pr.7] "Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription" | CHI 2025 | Yokohama, Japan
[Pr.6] "A Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians" | Bangladesh Association of Software Information Services (BASIS) National ICT Awards | 2022 | Dhaka, Bangladesh
[Pr.5] "Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians" | International Conference on 4IR (4th Industrial Revolution) for the Emerging Future | 2022 | Dhaka, Bangladesh
[Pr.4] "COVID-19 Defense System" | IEEE YESIST12 Maker Fair | 2021 | Dhaka, Bangladesh
[Pr.3] "Mobile App for Post COVID-19 Measures" | NASA Space Apps Challenge | 2021 | Dhaka, Bangladesh
[Pr.2] "Alcoholism Detection from 2D Transformed EEG Signal" | International Joint Conference on Advances in Computational Intelligence | 2020 | Daffodil International University, Dhaka, Bangladesh
[Pr.1] "EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data" | IEEE TENSYP2020 | Dhaka, Bangladesh

Awards & Scholarship

- [A.2] **EECS PBGA Summer Fellowship** (2024, 2025, 2026), UC Merced
[A.1] **Winner Award** (2022), Bangladesh Association of Software Information Services (BASIS) National ICT Awards, Dhaka, Bangladesh
[S.1] General Scholarship in Higher Secondary School Certificate (H.S.C.) examination (2015), Board of Intermediate and Secondary Education, Dhaka, Bangladesh

Core Skills

Research Methods: Human-subject experimental design, User studies & usability testing, Quantitative & qualitative data analysis, Statistical analysis

HCI & Interaction Design: Input technique design (gesture, gaze, force-based), Accessible/assistive interface design, Prototyping (low- and high-fidelity)

Brain-Computer Interfaces & Signal Processing: EEG signal acquisition & preprocessing, Motor imagery classification, Signal-to-image transformation techniques

Machine Learning & AI: TensorFlow, scikit-learn, Computer vision (object/gesture recognition)

Programming: Python, MATLAB, C++, Git/version control

Professional Affiliations

ACM ASSETS Program Committee Member	Since March, 2026
Health Sciences and Research Institute at UC Merced Student Member	Since November, 2024
Association for Computing Machinery (ACM) Student Member	Since September, 2023

References

Dr. Ahmed Arif <i>Associate Professor</i> Electrical Engineering and Computer Science University of California, Merced Email: asarif@ucmerced.edu Phone: 1 209-446-7292	Dr. Muhammad Aminul Haque Akhand <i>Professor</i> Computer Science and Engineering Khulna University of Engineering & Technology Email: akhand@cse.kuet.ac.bd Phone: 880 171-408-7205
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